COMP9315 23T1

Exercises 07

DBMS Implementation

Implementing Join: Nested-loop, Sort-merge, Hash

[Show with no answers] [Show with all answers]

1. Does the buffer replacement policy matter for sort-merge join? Explain your answer.

   [show answer]

2. Suppose that the relation *R* (with 150 pages) consists of one attribute *a* and *S* (with 90 pages) also consists of one attribute *a*. Determine the optimal join method for processing the following query:

   ```
   select * from R, S where R.a > S.a
   ```

   Assume there are 10 buffer pages available to process the query and there are no indexes available. Assume also that the DBMS only has available the following join methods: nested-loop, block nested loop and sort-merge. Determine the number of page I/Os required by each method to work out which is the cheapest.

   [show answer]

3. [Ramakrishnan, exercise 12.4] Consider the join $Join_{[R.a=S.b]}(R,S)$ and the following information about the relations to be joined:

   - Relation *R* contains 10,000 tuples and has 10 tuples per page
   - Relation *S* contains 2,000 tuples and also has 10 tuples per page
   - Attribute *b* of relation *S* is the primary key for *S*
   - Both of the relations are stored as simple heap files
   - Neither relation has any indexes built on it
   - There are 52 buffer pages available

   Unless otherwise noted, compute all costs as number of page I/Os, except that the cost of writing out the result should be uniformly ignored (it's the same for all methods).

   a. What is the cost of joining *R* and *S* using page-oriented simple nested loops join? What is the minimum number of buffer pages required for this cost to remain unchanged?

      [show answer]

   b. What is the cost of joining *R* and *S* using block nested loops join? What is the minimum number of buffer pages required for this cost to remain unchanged?

[show answer]

c. What is the cost of joining *R* and *S* using sort-merge join? What is the minimum number of buffer pages required for this cost to remain unchanged?

[show answer]

d. What is the cost of joining *R* and *S* using grace hash join? What is the minimum number of buffer pages required for this cost to remain unchanged?

[show answer]

e. What would be the lowest possible I/O cost for joining *R* and *S* using *any* join algorithm? How much buffer space would be needed to achieve this cost?

[show answer]

f. What is the maximum number of tuples that the join of *R* and *S* could produce, and how many pages would be required to store this result?

[show answer]

g. How would your answers to the above questions change if you are told that *R.a* is a foreign key that refers to *S.b*?

[show answer]

4. [Ramakrishnan, exercise 12.5] Consider the join of *R* and *S* described in the previous question:

a. With 52 buffer pages, if *unclustered* B+ tree indexes existed on *R.a* and *S.b*, would either provide a cheaper alternative for performing the join (e.g. using index nested loop join) than a block nested loops join? Explain.

   i. Would your answer change if only 5 buffer pages were available?
   ii. Would your answer change if *S* contained only 10 tuples instead of 2,000 tuples?

[show answer]

b. With 52 buffer pages, if *clustered* B+ tree indexes existed on *R.a* and *S.b*, would either provide a cheaper alternative for performing the join (e.g. using index nested loop join) than a block nested loops join? Explain.

   i. Would your answer change if only 5 buffer pages were available?
   ii. Would your answer change if *S* contained only 10 tuples instead of 2,000 tuples?

[show answer]

c. If only 15 buffers were available, what would be the cost of sort-merge join? What would be the cost of hash join?

[show answer]

d. If the size of *S* were increased to also be 10,000 tuples, but only 15 buffer pages were available, what would be the cost for sort-merge join? What would be the cost of hash join?

[show answer]

e. If the size of *S* were increased to also be 10,000 tuples, and 52 buffer pages were available, what would be the cost for sort-merge join? What would be the cost of hash join?

[show answer]

5. Consider performing the join:

```
select * from R, S where R.x = S.y
```

where we have $b_R = 1000$, $b_S = 5000$ and where *R* is used as the outer relation.

Compute the page I/O cost of performing this join using *hybrid hash join*:

a. if we have $N=100$ memory buffers available

[show answer]

b. if we have $N=512$ memory buffers available

[show answer]

c. if we have $N=1024$ memory buffers available

[show answer]